

EoI : CGIAR SCALING FOR IMPACT

Section 0 – Title and contact Information

- Title of the proposed solution track: *Closing the Adaptation Gap in agriculture: Scaling Agvisely Climate Services to Build Resilient Farming Systems in Bangladesh*
- Title of the AoW2 Flagship that the solution track aligns with: Flagship 1.1. Scaling digital climate services for farmers and enterprises.
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- Key CGIAR Center Partners (in the case of multi-Center Eols)

The CGIAR centers participating in the EOI are the International Maize and Wheat Improvement Center (CIMMYT, lead), the International Rice Research Institute (IRRI), and WorldFish.

- Technical link to other Science Programs – Climate Action Science Program, Sustainable Farming Science Program, SAAF Science Program, Digital transformation accelerator.

Section 1 – Innovation Package, Value Proposition & Alignment

Purpose: The innovation selected for scaling is Agvisely (agvisely.com), a climate service developed by the International Maize and Wheat Improvement Center (CIMMYT), WorldFish, the Department of Agricultural Extension (DAE), the Bangladesh Livestock Research Institute (BLRI), and the Bangladesh Meteorological Department (BMD). Endorsed by the

Ministry of Agriculture in 2019, Agvisely is an automated advisory system that integrates BMD's Weather Research and Forecasting (WRF) model to produce localized forecasts every three hours for all 495 sub-districts (Upazilas). It links weather data to species-specific thresholds for crops, livestock, and aquaculture across major phenological and growth stages. Automation ensures reliable advisories without manual oversight. Currently, Agvisely reaches 587,474 farmers (PRMS report, appendix). The next phase aims to scale out to five million users and scale deep by embedding within DAE, BLRI, Department of Livestock (DLS), Bangladesh Fisheries Research Institute (BFRI), Department of Fisheries (DoF), and Bangladesh Agricultural Research Council (BARC), while expanding API access for public and private services.

The problem to be solved: Bangladesh's smallholder farmers face intensifying climate threats—erratic rainfall, temperature extremes, cyclones, floods, droughts, and salinity—disrupting crops, livestock, and aquaculture. Yet advisory systems remain fragmented, manual, poorly localized, and insufficiently inclusive, leaving most farmers without timely, actionable guidance. To close this gap, scaling an automated climate service like Agvisely that delivers localized forecasts and advice, enabling farmers to adapt practices, reduce losses, safeguard livelihoods, and build resilience—while ensuring inclusion of women, youth, and marginalized groups, is critical. Agvisely covers ten major vulnerable crops: cereals -rice, wheat, maize (77% of cropped area); lentil and mungbean (51% area under pulses); mustard (66% area of oilseeds); and potato (50% of vegetables). For aquaculture, Rohu, Tilapia, and carps (>50% production), as well as shrimp/prawn (74% fishery export earnings). In livestock, cattle are financial assets of value of \$376 per household, while poultry are worth \$176; weather advisories are key to protecting these assets.

The scaling ambition:

The ambition is to significantly scale Agvisely, which currently reaches ~500,000 farmers each season, to 5 million farmers by 2028–2030, with at least 30% of them actively using the advisories. Scaling activities cover crops, livestock, and aquaculture—working through extension agents, lead farmers, financing agencies, and other value chain actors. Rice requires particular emphasis, as 15 million rice farmers in the country are vulnerable to climate stress, while 1.8 million aquaculture farmers face similar risks. More than 80% of farmers keep cattle or poultry, collectively holding 20 million cattle and 327 million chickens.

By scaling both out (expanding reach) and deep (embedding within government systems), this track will contribute to S4I's KPIs on reach, institutionalization, and climate resilience. However, projected reach is contingent upon the expected CGIAR W2 investment from the

UAE Government through the Agriculture Innovation Mechanism for Scale (AIM4Scale), alongside confirmed funding from CIMMYT's bilateral CSISA-MEA project.

- **Innovation & Value Proposition**

Agvisely is an automated, climate-risk–threshold advisory system for the farming system that downscales forecasts and combines them with crop, aquaculture and livestock-specific biological- thresholds to deliver actionable advisories across all 495 sub-districts. The platform's value lies in precise, timely, and accessible localized advisories validated by the national agricultural research system. In addition, it provides stage-specific good agricultural. livestock/aquaculture practices and integrated pest management advisory. Benefits include reduced risks, increased productivity, resilience and nutrition. Rice, aquaculture and livestock cover 6%,4% and 2% of GDP. Saving 1% of the cereal production in Bangladesh (62-66 million tonnes) meet annual calorific requirements of 2.8-3 million people, 1% saving in aquacultural production provide 24,000 tons that can meet annual protein needs of one million people (60 g/day/person). Similarly, if 1% of milk production is saved, it can meet protein requirement of 0.2 million people. This clearly demonstrates the value proposition of Agvisely.

- **Customers & Alignment**

Institutional customers include the Department of Agricultural Extension (DAE), Agricultural Information Service (AIS), Department of Livestock Services (DLS), Department of Fisheries (DoF), Bangladesh Agricultural Research council (BARC-Khamari app) and Nongovernmental organizations (NGOs), financial (eg: PKSf) and insurance providers. Private sector actors include agribusinesses, financing agencies, insurance providers, digital agricultural application developers and other value chain actors. Other potential customers are cooperatives and lead farmers. End-users are smallholder farmers, including women and marginalized groups. The track aligns directly with Flagship 1.1. Scaling digital climate services for farmers and enterprises.

- **Evidence of Impact** – Provide citations and evidence of both effectiveness (impact) and scalability in practice (uptake beyond pilots, adoption drivers). (~150 words)

Agvisely is only scaled through the agricultural extension pathway as of now. It reaches 587,474 farmers through 8,000 registered extension officers, with endorsements from the Ministry of Agriculture since 2019. MEL data shows that farmers using Agvisely advisories adapt management practices more effectively to weather risks. The PRMS report is attached with the optional Annex section. The Department of Agricultural Extension (DAE)

has issued a circular mandating their extension officers to use the Agvisely application. (Note that DAE has more than 14,092 extension officers, of which, Agvisely has 8000 registered officers). In 2026-2028, Agvisely is expected to scale through multiple public and private extension partners, bundled service models, NGOs and cooperatives, AI innovations creating voice enabled services for inclusive service and strategically aligning with large scale investments by multilateral agencies (World Bank, ADB, IFAD etc.)

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

- **Purpose:**

1) Scaling out: The target is to expand Agvisely’s reach from its current base of 587,474 farmers to at least five million (30% use). **(2) Scaling deep:** Embed Agvisely systematically within the annual workplans and activities of AIS, DAE, DoF, DLS to institutionalize its’ use, securing long-term sustainability. At the same time, Agvisely’s API will be made available to private extension and finance service providers. **(3) Capacity building:** At least 4,000 additional DAE/DLS/DOF extension staff, lead farmers, private sector and NGO actors will be trained to mainstream Agvisely in their advisory services reinforcing institutional ownership. **(4) Bundled service models:** The integration of Agvisely advisories into financial and service bundles including crop and fish insurance products and loan services. **(5) Inclusive AI innovation:** Prototype and deploy AI systems to generate text to voice in Bangla, including regional dialects, boosting access for low-literacy farmers. **(6) Scaling through NGO partnerships.** NGOs like BRAC, CNRS, ASA to use Agvisely through existing programmes or projects **(7) Programmatic scaling:** Link Agvisely with payment for results-based lending programs of IFAD and the World Bank (specifically the Program on Agricultural and Rural Transformation for Nutrition, Employment, and Resilience ([PARTNER](#)) project), and potentially to ADB programs under development in Bangladesh to achieve scale. Between 2026 and 2028, Agvisely will reach five million end-users—1.5 million through public extension channels, 1.5 million via private extension, finance and API services, 0.5 million through NGO partnerships, and 1.5 million through large-scale programmatic platforms—ensuring both breadth and depth of impact across farming communities.

- **Partnerships & Leverage**

In case of crop agriculture, CIMMYT collaborates with BARI and DAE as key partners. The partners for the livestock service are Bangladesh Livestock Research Institute (BLRI) and Department of Livestock services (DLS). CIMMYT, WorldFish, Department of Fisheries (DoF) and Bangladesh Fisheries Research Institute (BFRI) collaborated in the development of aquaculture climate services. For the meteorological information, the partners are Bangladesh meteorological Department (BMD) and Regional Integrated Multi-Hazard Early Warning System for Africa and Asia (RIMES). In the case of public sector actors, the DAE has

issued a circular mandating the extension officers to use Agvisely in their routine work schedule, which is evidence of institutionalization (attached in annex). CIMMYT is exploring the collaboration with Bangladesh Agricultural Research Council (BARC) and Agricultural information Service (AIS) to integrate Agvisely in their platforms (eg: Khamari app of BARC and digital services of AIS). WorldFish is exploring the integration of Agvisely in the OneFish app. In the case of private companies, IFarmer (0.26 million farmers in the platform) and Genius Farms expressed interest for integrating Agvisely API in their application (see annex for the letters for expressing demand). In the case of financial institutions, PKSF (the apex body of microfinance institutions) and Green Delta insurance, have expressed their interest in scaling Agvisely in their programs. CIMMYT has MOU with BARC, BARI, BMD and DAE. WorldFish has MOU with DoF and BFRI. In survey regarding demand for Agvisely's livestock services, Adorsho PraniSheba Ltd, iFarmer Ltd, InsureCow Ltd and Digital Innovation for Impact Ltd. expressed interest in integrating the services in their insurance and credit related programs. In the case of aquaculture service, Adorsho praniSheba Ltd, ifarmer Ltd, Digital Innovation for Impact Ltd and Shushilan NGO have expressed interest. In case of co-investments. CGIAR W2 investment from UAE Govt. through the Agriculture Innovation Mechanism for Scale (AIM4Scale) during 2026-2028 and the confirmed funding from the bilateral project of CSISA-MEA project of CIMMYT.

- **Risks & Bottlenecks**

The implementation of Agvisely climate services in Bangladesh faces potential risks that need to be carefully managed. **Institutional risks** may arise due to limited capacity or shifting priorities within key public sector partners. To mitigate this, Agvisely should be embedded in institutional workplans and complemented with continuous staff training to ensure ownership and sustainability. **Financial risks** are also significant, particularly regarding the long-term sustainability of the service once donor funding diminishes. A practical mitigation strategy is to design bundled financial services and explore cost-recovery models, such as providing private sector access to APIs. On the **operational** side, gaps in farmers' digital literacy could limit uptake. Addressing this challenge requires testing AI-driven, voice-based advisories in Bangla and local dialects. Finally, **social risks** exist in the form of gender barriers, which can constrain women's access and use of digital advisory services. This can be mitigated by setting explicit inclusion targets, such as ensuring that at least 35% of users are women, and by promoting women-to-women service provider models that enhance trust and accessibility, especially through NGO partners.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

- **Readiness & Responsible Scaling**

Agvisely is at an advanced scaling readiness stage, with proven uptake (>0.5 million reached), government endorsement, and API integration. Scaling will be responsible and inclusive, with explicit gender targets, accessible voice advisories, and participatory co-design with local institutions. Environmental safeguards are built-in through climate-smart advisory content. A scaling scan assessment was conducted in 2023 for Agvisely and the recommendations are already integrated (the report is in the Annex). The study on digital service provider models for women for Agvisely that is a pathway for inclusion is attached in the annex.

- **Achievable target, Resources & Activities**

1) 2026 Months 0–6 –agreements formalized with DAE, DLS, AIS, BARC and the [Institute of Water Modeling](#) (IWM; which manages the back-end of the Khamari app) to integrate dynamic advisories from Agvisely; **(2) 2026 Months 6–12** – Agvisely’s API integrated into BAMIS, managed by DAE, which currently produces biweekly, district-level advisories written manually by a committee. By contrast, Agvisely automatically and dynamically assembles and distributes localized advisories, thereby adding precision and timeliness to BAMIS outputs. Integration will also extend to Khamari and AIS call center/SMS systems, enabling a unified delivery architecture; Agvisely’s API integrated into private sector applications **(3) 2026 Months 6-12 2027 months 0-6** – at least 4,000 extension officers, lead farmers, and private extension company staff trained utilize and disseminate information derived from Agvisely; Agvisely’s API integrated World Fish’s OneFish app **(4) 2026 Months 1–12 and 2027 Months 1-12** – Multiple agreements are envisioned to be signed with financial or insurance institutions to bundle Agvisely advisories with loans, insurance, for crop, fish and livestock services; **(5) 2026 Months 1–12 and 2027 Months 1–12** – AI-enabled voice prototype operational in Bangla via hotline services, likely offered by AIS, to improve accessibility for low-literacy farmers; **(6) 2026 Months 6-12 and 2027 Months 1-12** - Engagement of NGOs to scale agvisely through their programmes **(7) 2028 Months 1-12** – further scaling to five million farmers using information derived from Agvisely to support climate-resilient farm management, with Agvisely embedded in public advisory systems and aligned with major donor programs, including the World Bank, IFAD, and ADB.

CIMMYT has three international staff and 10 trained locally hired staff committing different levels of FTEs to support the scaling activities mentioned in the next section. IRRI offers two international staff and 5 locally hired staff committing different levels of FTEs. WorldFish has two international staff and 3 locally hired staff to support the scaling activities. Potential CGIAR W2 investment from UAE Govt. through the Agriculture Innovation Mechanism for Scale (AIM for Scale) during 2026-2028 and the confirmed funding from the bilateral project of CSISA-MEA project of CIMMYT is leveraged here.

- **Timeline, Milestones & Monitoring**

A robust MEL framework will be established to track Agvisely's progress toward reaching five million users and ensuring institutional ownership, quality delivery, and farmer uptake. The plan will combine process monitoring, outcome tracking, and adaptive learning to adjust interventions as needed. Key activities and timelines include:

2026 (Months 0–6): Secure agreements with DAE, DLS, and AIS for co-branded delivery of Agvisely advisories, ensuring government ownership. Integrate private sector partners providing water temperature data into Agvisely-aquaculture.

2026 (Months 0–12): Embed Agvisely's API into BAMIS or Khamari applications; link with AIS's call centers; extend API access to nonprofit and for-profit service providers; link to WorldFish's OneFish app; secure agreement with DoF.

2027 (Months 0–6): Develop and deliver training modules for 4,000 extension staff, lead farmers, service providers, input dealers, NGOs, and private companies on using and disseminating Agvisely advisories.

2026–2027 (Months 0–12): Pilot integration of Agvisely advisories with financial and insurance products; secure NGO agreements for scaling Agvisely within their programs.

2026–2027 (Months 0–12): Prototype AI-enabled advisories in Bangla and regional dialects tailored to specific farmer profiles.

2026 (Months 6–12) and 2027–2028 (Months 0–12): Roll out farmer enrollment and outreach campaigns via extension networks, farmers' groups, and agribusiness partners.

2027–2028 (Months 0–12): Expand Agvisely's user base from 500,000 to at least five million farmers, leveraging integration with IFAD, World Bank, and ADB investment programs..

Optional Annexes

- Scaling Scan result: <https://cgspace.cgiar.org/items/5a66383d-1bd9-4f18-a6ec-418b07b894db> (Note that the assessment is conducted in 2023 and most of the recommendations are completed in 2025)
- MELIA study: PRMS report of Agvisely
- **Publications:**
 - 1) Agvisely livestock – user manual: <https://cgspace.cgiar.org/bitstreams/ab848ee2-d789-4a2d-a91f-8eb3ba6a5ceb/download>
 - 2) Agvisley crops – project report- <https://cgspace.cgiar.org/items/d275a2b0-b09e-44b2-80ed-78564edd450a>

- 3) Agvisely crops - Women to women service provider model for Agvisely: <https://cgspace.cgiar.org/items/d2d21bff-b80c-4de1-86bd-5922cd68520b>
 - 4) Agvisely aquaculture - <https://www.frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2021.677069/full>
 - 5) Aquaculture climate service: <https://hdl.handle.net/20.500.12348/5750>
- **Partner commitments**
 - A) Circular issued by Department of Agricultural Extension (DAE) mandating the use of Agvisely for extension officers.
 - B) Letter of demand – iFarmer Ltd. – Agricultural finance and digital services company (0.26 million farmers)
 - C) Letter of demand – Genius Farms Ltd. – AI based pest and disease detection service